CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a cyclic quadrilateral with sides 5 cm, 7 cm, 15 cm and 19 cm is (cm^2): 1

(a) 95 **(b)** 98 **(c)** 96 **(d)** 97

2. A circle of radius 7 cm (take $\pi = \frac{22}{7}$) is inscribed in a square. The area of the square not covered by the circle is (cm^2): 1

(a) 42 **(b)** 43 **(c)** 44 **(d)** 41

3. Two concentric circles have radii 14 cm and 21 cm (take $\pi = \frac{22}{7}$). The area of the ring between them is (cm^2): 1

(a) 771 **(b)** 769 **(c)** 772 **(d)** 770

4. A wheel makes 100 revolutions to cover 220 m (take $\pi = \frac{22}{7}$). Its radius (cm) is: 1

(a) 37 **(b)** 35 **(c)** 36 **(d)** 34

5. The area of a rhombus of side 13 cm and one diagonal 10 cm is (cm^2): 1

(a) 121 **(b)** 119 **(c)** 120 **(d)** 122

6. A triangle has sides 10 cm, 17 cm, 21 cm. The length of the shortest altitude (cm) is: 1

(a) 10 **(b)** 9 **(c)** 7 **(d)** 8

7. A rope of length 88 cm (take $\pi = \frac{22}{7}$) forms a circle. The same rope forms a square. The area of the square (cm^2) is: 1

(a) 486 **(b)** 484 **(c)** 485 **(d)** 483

8. If the radius of a circle is doubled, its area becomes: 1

(a) two times **(b)** four times **(c)** eight times **(d)** three times

9. **Assertion (A):** The area of a triangle with sides 13 cm, 14 cm and 15 cm is 84 cm^2. 1
Reason (R): By Heron's formula the area is $\sqrt{s(s-a)(s-b)(s-c)}$, where s is the semi-perimeter.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The circumference of a circle of radius 7 cm is 44 cm (take $\pi = \frac{22}{7}$). 1
Reason (R): The area of a circle of radius r is πr^2.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the area of a cyclic quadrilateral with sides 5, 7, 15 and 19 cm. **2**

12. A circle of radius 7 cm is inscribed in a square (take $\pi = \frac{22}{7}$). Find the area of the square left uncovered by the circle. **2**

13. The area of a rhombus is 120 cm^2 and one diagonal is 10 cm. Find the other diagonal and the side of the rhombus. **2**

14. Find the area of a cyclic quadrilateral with sides 5, 8, 19 and 20 cm. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rope 44 cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. **3**

16. A running track is made of a rectangle 60 m $\times$ 28 m with a semicircle (radius 14 m) on each of the two shorter sides (take $\pi = \frac{22}{7}$). Find the length of the track along its boundary and the area enclosed. **3**

17. The sides of a triangle are 13 cm, 14 cm and 15 cm. Find its area and the altitudes on the sides 13 cm and 14 cm. **3**

OR

A rope 88 cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. **3**

18. A running track is made of a rectangle 80 m $\times$ 14 m with a semicircle (radius 7 m) on each of the two shorter sides (take $\pi = \frac{22}{7}$). Find the length of the track along its boundary and the area enclosed. **3**

OR

The sides of a triangle are 15 cm, 20 cm and 25 cm. Find its area and the altitudes on the sides 15 cm and 20 cm. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A circular pond of radius 14 m in a village in Kerala is surrounded by a 7 m wide path. **5**

Take $\pi = \dfrac{22}{7}$.

(i) Find the circumference of the pond.

(ii) Find the area of the pond.

(iii) Find the radius and the area of the outer circle formed by the path.

(iv) Find the area of the path and the cost of paving it at Rs 40 per m^2.

20. Two concentric circles have radii 21 cm and 28 cm (take $\pi = \frac{22}{7}$). Find the area of the ring and the difference between the two circumferences. **5**

OR

In quadrilateral $ABCD$, $AB = 5$ cm, $BC = 12$ cm, $CD = 14$ cm, $DA = 15$ cm and $\angle ABC = 90^\circ$. Find the area of $ABCD$. **5**

END OF QUESTION PAPER — Total: 40 Marks